Supernova Astrophysics with Gravitational Waves: High School Lesson Plan for Astrostatistics

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Course Overview

	Topic	Example System	Astrophysics Content	Coding Skill
1	Intro to GWs and LIGO	Compact Binary Mergers	GW basics, spacetime ripples	Python crash course
2	Binary Systems & GW Generation	Black Hole Binary	Orbit dynamics, inspiral	PN waveform modeling
3	Merger & Ringdown	Same	NR merger physics	Ringdown spectrum
4	Data Analysis	LIGO BBH Event	Signal processing	LIGO data filtering
5	Multi-Messenger Astronomy	Binary Neutron Stars	EM + GW signals	Parameter estimation
6	Core-Collapse Supernovae	Supernova Explosions	Collapse dynamics	PCA waveform modeling
7	Supernovae & LIGO	SN GW Signatures	Non-detections	Time-frequency search
8	Inference & Projects	Final Project	Formation channels	Inference toolkit

We will work in the textbook by R. D'Inverno and J. Vickers: *Introducing Einstein's Relativity: A Deeper Understanding*. For extra material, see *Spacetime Physics* by Taylor and Wheeler.

An overview of the concepts and questions that you will be able to answer by the end:

1. Newtonian gravity
   (a) Newton's three laws of motion
   (b) law of gravitation
   (c) Laplace equation and Newton's second law (specify the field, then compute the consequences for a test mass)
   (d) visualizing solutions with Python
   (e) thought experiments for black holes and gravitational waves
2. Einsteinian gravity
   (a) detour to special relativity: the case of high speed and no gravity
   (b) Einstein's predecessor's and motivations for a new theory of gravity
   (c) demystifying tensors and geodesics (with quick detour to math history of curved geometry)
   (d) the Einstein field equations and the geodesic equation
   (e) the nonrotating (Schwarzschild) and rotating (Kerr) spacetimes
3. Black holes and gravitational waves
   (a) observations: the BH mass spectrum and the GW freq/wavelength spectrum -¿ theory: what do we learn from the observations?... eg's astrophysics, cosmology, and the quantum physics in neutron stars (and black holes!).

Example System 1: Compact Binary Mergers

Week 1: Introduction to Gravitational Waves

Astrophysics Topics:

- What are gravitational waves (GWs)?
- GW sources: black holes, neutron stars
- LIGO, Virgo, and KAGRA detectors

Source: Abbott et al. (2016), "Observation of Gravitational Waves from a Binary Black Hole Merger," Phys. Rev. Lett.

Python Module: Plotting a Toy GW Signal

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 1, 1000)
h = np.sin(2 * np.pi * 100 * t) * np.exp(-100 * t)

plt.plot(t, h)
plt.xlabel("Time (s)")
plt.ylabel("Strain")
plt.title("Toy GW Signal")
plt.grid(True)
plt.show()
```

Week 2: Binary Systems and PN Inspiral

Astrophysics Topics:

- Binary orbital dynamics
- Gravitational wave emission
- Post-Newtonian (PN) approximations

Source: Blanchet (2014), Living Reviews in Relativity

Python Module: Inspiral Frequency Evolution

```
from scipy.integrate import solve_ivp
import numpy as np
import matplotlib.pyplot as plt

def dfdt(t, f, M_chirp):
    G = 6.6743e-11
    c = 3e8
    return (96/5) * np.pi**(8/3) * (G * M_chirp / c**3)**(5/3) * f**(11/3)

M_chirp = 1.2 * 1.989e30

sol = solve_ivp(dfdt, [0, 10], [20], args=(M_chirp,), t_eval=np.linspace(0, 10,
   1000))
plt.plot(sol.t, sol.y[0])
```

```
plt.xlabel("Time (s)")
plt.ylabel("Frequency (Hz)")
plt.title("GW Frequency Chirp")
plt.show()
```

Week 3: Merger and Ringdown

Astrophysics Topics:

- Numerical Relativity (NR) merger phase
- Ringdown and quasi-normal modes

Source: Berti et al. (2009), CQG

Python Module: Ringdown Waveform (Damped Sinusoid)

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 0.1, 1000)
h = np.exp(-300 * t) * np.cos(2 * np.pi * 1500 * t)

plt.plot(t, h)
plt.title("Ringdown GW Signal")
plt.xlabel("Time (s)")
plt.ylabel("Strain")
plt.grid(True)
plt.show()
```

Week 4: LIGO Data Analysis

Astrophysics Topics:

- GWOSC and open data
- Black hole merger event GW150914

Python Module: GWOSC Data Access using `gwpy`

```
from gwpy.timeseries import TimeSeries

strain = TimeSeries.fetch_open_data('H1', 1126259462, 1126259474)
strain.plot()
```

Week 5: Binary Neutron Stars and Multi-Messenger Events

Astrophysics Topics:

- BNS merger physics and kilonovae
- GW170817 and gamma-ray burst counterpart

Source: Abbott et al. (2017), GW170817 multi-messenger discovery

Python Module: Posterior Visualization with `corner.py`

```
import corner
import numpy as np

samples = np.random.multivariate_normal([1.4, 0.1], [[0.01, 0.002], [0.002,
   0.001]], size=1000)
corner.corner(samples, labels=["Mass (M)", "Tidal deformability"])
```

Example System 2: Supernova Explosions

Week 6: Supernova Models and GW Signals

Astrophysics Topics:

- Core-collapse mechanisms (neutrino-driven vs magnetorotational)
- GW emission and modeling uncertainties

Sources: Andresen et al. (2017), Powell et al. (2019)

Python Module: PCA of Supernova GW Waveforms

```
import numpy as np
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

waveforms = np.random.randn(100, 500)
pca = PCA(n_components=3)
components = pca.fit_transform(waveforms)
reconstruction = pca.inverse_transform(components)

plt.plot(reconstruction[0])
plt.title("Reconstructed Supernova GW Signal")
plt.xlabel("Time steps")
plt.ylabel("Strain")
plt.show()
```

Week 7: GW Searches for Supernovae in LIGO

Astrophysics Topics:

- Why hasn't LIGO detected SN GWs?
- Q-transform and time-frequency searches

Python Module: Q-transform of Simulated Data

```
from gwpy.timeseries import TimeSeries
from gwpy.signal import filter_design
import numpy as np

data = TimeSeries(np.random.randn(4096), dt=1/4096)
white = data.whiten()
white.q_transform(outseg=(0, 2)).plot()
```

Week 8: Inference and Final Projects

Project Ideas:

1. Model your own BBH or SN waveform
2. Use LIGO data to detect a real event
3. Compare SN GW models with PCA
4. Build a multi-messenger timeline of GW170817

Resources

1. GWOSC Tutorials
2. Blanchet (2014), Living Reviews in Relativity
3. Berti et al. (2009), CQG
4. GW150914 Open Data
5. Supernova GW Model Archive